

WHATI, NT, Canada

WMO: 711630

Lat: 63.134N

Lon: 117.244W

Elev: 890

StdP: 14.23

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.7	-38.0	-49.1	0.3	-40.7	-45.3	0.4	-37.7	16.0	-0.1	14.3	-4.5	1.6	270	0.741

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	81.1	61.4	77.8	60.2	74.4	59.0	63.8	76.0	62.2	74.0	60.7	71.8	8.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.6	78.6	67.6	57.6	73.2	65.8	55.8	68.5	64.6	29.6	76.3	28.4	74.0	27.3	71.9	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	14.5	13.1	-45.5	86.4	5.6	3.4	-49.5	88.8	-52.8	90.8	-56.0	92.8	-60.1	95.2		
(5)			-44.9	67.0	5.2	1.3	-48.7	67.9	-51.7	68.6	-54.7	69.3	-58.5	70.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.0	-13.1	-8.2	0.8	21.5	41.9	56.3	61.5	55.7	44.2	27.7	6.5	-8.4		
	DBStd	28.93	14.26	12.86	14.93	13.30	10.55	7.90	6.42	6.86	7.77	8.35	12.40	14.94		
	HDD50	10323	1958	1629	1524	856	289	30	3	24	205	691	1305	1809		
	HDD65	15013	2423	2049	1989	1306	717	277	145	299	623	1156	1755	2274		
	CDD50	846	0	0	0	0	38	218	358	200	32	0	0	0		
	CDD65	60	0	0	0	0	0	15	36	9	0	0	0	0		
	CDH74	799	0	0	0	0	24	216	407	147	5	0	0	0		
	CDH80	146	0	0	0	0	1	43	76	25	1	0	0	0		
	WSAvg	6.2	4.6	5.1	6.1	6.8	7.4	7.4	6.9	7.0	7.2	6.6	5.3	3.8		
	PrecAvg	11.7	0.5	0.4	0.5	0.4	0.7	1.4	1.9	2.0	1.5	1.1	0.7	0.6		
(15) Precipitation	PrecMax	13.9	1.0	0.8	1.1	0.6	1.4	3.7	3.7	3.2	3.1	1.7	1.5	1.1		
	PrecMin	8.8	0.2	0.2	0.1	0.2	0.2	0.3	1.0	0.7	0.4	0.4	0.4	0.2		
	PrecStd	1.4	0.2	0.2	0.2	0.2	0.4	0.7	0.7	0.6	0.3	0.2	0.2	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.4	28.3	44.1	57.1	77.6	85.2	86.1	83.0	71.8	52.5	33.3	27.8		
		MCWB	27.4	26.1	36.5	44.4	55.0	61.6	64.1	62.6	57.4	47.0	32.4	26.7		
	2%	DB	20.7	19.6	36.5	51.0	70.7	79.9	81.8	78.0	65.9	45.6	29.8	21.8		
		MCWB	19.9	18.4	31.7	40.8	52.8	59.2	62.3	61.1	54.0	41.4	29.0	21.3		
	5%	DB	11.7	15.0	30.8	46.7	65.4	75.8	78.5	73.8	61.8	41.4	26.3	17.3		
		MCWB	11.2	14.1	27.7	38.3	50.3	57.9	61.4	59.9	52.5	39.1	25.6	16.7		
	10%	DB	5.9	10.8	24.9	41.9	59.7	71.7	75.3	69.8	57.5	38.2	23.2	12.5		
		MCWB	5.4	10.0	22.7	35.5	47.3	56.2	60.6	58.1	50.7	36.5	22.6	11.9		
	0.4%	WB	27.4	26.3	36.9	44.4	56.4	63.7	66.6	64.9	59.0	46.9	32.7	26.7		
		MCDB	28.5	28.6	43.2	55.9	72.8	80.4	78.0	77.6	67.7	51.6	33.9	27.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	20.0	18.4	32.0	41.2	53.2	61.3	64.6	62.9	56.0	42.2	29.2	21.2		
		MCDB	20.9	19.6	35.8	50.3	68.6	75.4	76.0	74.2	63.2	44.8	29.8	21.7		
	5%	WB	11.2	14.1	27.6	38.2	50.6	59.5	63.1	61.0	53.5	39.1	25.6	16.9		
		MCDB	11.7	14.9	30.7	45.5	64.0	72.3	74.8	71.3	60.2	41.0	26.2	17.4		
	10%	WB	5.4	10.1	22.5	35.5	47.8	57.6	61.7	59.2	51.2	36.6	22.7	12.0		
		MCDB	5.9	10.8	24.8	41.2	59.1	69.3	73.0	67.9	57.1	38.2	23.1	12.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	18.9	25.0	25.1	21.0	20.3	20.3	19.4	16.4	10.6	13.8	14.9		
		MCDBR	16.8	21.6	28.2	27.6	28.7	26.4	26.1	27.3	24.4	16.1	13.1	17.8		
		MCWBR	16.7	20.4	23.1	18.6	14.4	11.4	11.4	13.4	14.3	12.0	12.6	17.5		
		MCDBR	16.6	21.4	27.4	25.6	27.2	23.3	22.3	22.8	21.7	14.5	12.4	17.7		
	5% WB	MCWBR	16.4	20.3	22.5	17.5	14.3	10.7	10.7	12.5	13.7	11.0	12.0	17.5		
(40) Clear-Sky Solar Irradiance	taub		0.180	0.228	0.257	0.298	0.327	0.338	0.369	0.335	0.304	0.262	0.177	0.149		
	taud		2.017	2.262	2.268	2.254	2.445	2.467	2.394	2.528	2.569	2.453	2.035	1.943		
	Ebn at Noon		156	239	273	281	282	281	266	268	257	225	154	100		
	Edh at Noon		9	17	26	34	31	31	32	26	20	15	9	4		
(44) All-Sky Solar Radiation	RadAvg		82	347	924	1588	1853	1928	1682	1270	761	282	88	35		
	RadStd		6	18	43	82	124	188	89	84	72	34	14	2		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	+2.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page